

Corroboration Is All You Can Certify

The Minimum Cut as the Limit of Certified Defenses for GraphRAG

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Abstract—Artifact. Code, data and a one-command reproduction gate: <https://anonymous.4open.science/r/graphrag-certified>.

A GraphRAG system answers a multi-hop question by retrieving a set of related facts and reasoning over them. Certified defenses against poisoning split that evidence into isolated cells, ask each cell for a verdict, and vote, on the assumption that one poisoned item spoils one cell. Relational evidence breaks the assumption. A cell can check a pattern only if it holds every relation in the pattern, so the same relation must be copied into many cells, and one forged relation then corrupts every copy it reaches. Isolation and exposure turn out to be the same lever.

We first ask how much that copying has to cost. For defenses that place evidence without looking at it, a deterministic partition achieves a replication exponent set by the anchored fractional edge packing number of the query while still detecting every pattern, and a matching lower bound holds wherever the integral and fractional packing numbers agree, which they do on all 35 query families and benchmark queries we evaluate. The packing number is at most the number of relations in the query, and equals it only in one special case, so the rate assumed by earlier certificates charges too much: on the 22 TPC-H queries it over-prices 18 of the 20 multi-relation ones, by a median $3.2\times$ and up to $32\times$.

We then show that replication is not what certified robustness is made of. What limits any certificate is how much independent support the retrieved evidence contains, measured as the fewest relations that meet every piece of supporting evidence. We prove that no defense of any architecture certifies more than half of that number against an adversary that can remove support, or one less than all of it against an adversary that can only add, and that both bounds are attained. The factor two is not the price of a fallible reader: it is charged exactly when adversarial text can remove support, by modifying relations or by sharing a reader’s context with honest evidence, which every relation-level partition allows. Isolating facts rather than relations removes it. For multi-hop questions the quantity is a minimum edge cut, so one max-flow gives the certificate for each answer, and the same cut is the cheapest attack that rewrites relations.

We check the theory on every connected query pattern up to five edges, the 22 TPC-H queries, two knowledge-graph workloads, a public power-network topology and five implemented adversaries, with a gate that asserts every number in

the paper against committed data, and run fact isolation end to end with an LLM reader against shared-relation forgery, knowledge-evolution modification, a min-cut adaptive attacker and prompt injection: no attack within the certified budget changed a report in 864 attacked runs, and injection never succeeded. On MetaQA 96% of two-hop answers rest on a single chain, while 72% of Hetionet’s drug–disease claims, checked against a declared class of seven metapaths, certify at least one forged relation against 12% of non-claims: certifiability is set by the support class a deployment declares. The conclusion for practice is that certifiable robustness is governed by independent relational corroboration, not by the number of cells a defense can afford.

1. Introduction

Retrieval-augmented generation [30] grounds a language model in documents fetched at query time; GraphRAG [31] replaces the flat document store with a graph of entities and relations, and a family of graph-structured variants has followed it [32]. A GraphRAG system answers a multi-hop question by retrieving a subgraph around an entity and reasoning over it: *which drug interacts with a target this protein regulates, does this claim trace back to a primary source, is this account on a cycle of transfers*. What these share is not a domain but a shape: the answer is a *relational pattern* that no single retrieved fact establishes alone. Recent work shows such systems are poisoned in a specific way, not by forging the conclusion but by forging a single *relation*. One fabricated edge participates in many candidate pieces of evidence at once, so one injection propagates [10], [4].

The mechanism those attacks exploit is *shared* relational structure: one forged relation propagates because many candidate witnesses reuse it. This paper studies the mirror image, that *independent* relational structure is a defensive resource, and asks how much of it the retrieved evidence contains. The two are different views of the same underlying combinatorial object, which is why a single linear program can characterize both.

The standard defense is architectural and familiar from data poisoning: split the retrieved evidence into K isolated cells, ask an independent verdict of each, and take a majority vote [9], [21]. Isolation is supposed to bound the blast radius. On *passages* it does, because a passage means something on its own and the groups can be disjoint. On relations it does not.

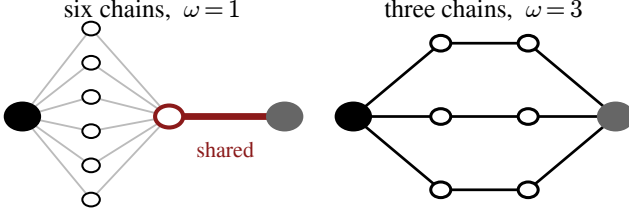


Figure 1. Numerosity is not independence. Left: six candidate chains that all run through one relation are a single point of failure, and the certificate correctly reports $\omega = 1$. Right: three chains that share no relation are three witnesses. Only the right-hand evidence can be certified, and the difference is invisible to any count of chains.

The problem. A cell can only recognise the pattern if it holds the *whole* pattern, so the same edge must be copied into many cells. Call that copy count ρ . Every copy is also a place the adversary lands: forging one edge corrupts all ρ of its cells. Detection forces replication, and replication is the adversary’s amplification.

The limit. Strip the architecture away and ask what any defense can certify. Write τ for the fewest retrieved relations that meet every witness: how much support the adversary would have to take away to leave none. No defense, of any architecture and at any compute, certifies more than $\lceil \tau/2 \rceil - 1$ forged relations against an adversary that can remove support, or $\tau - 1$ against one that can only add, and both bounds are attained (Theorem 5.4). For a multi-hop question τ is a minimum cut and equals the number of edge-disjoint supporting chains, so the certificate is one max-flow (Corollary 6.1), and the same cut is the cheapest attack that rewrites relations. Figure 1 is the whole idea in one picture: chains that share a relation share a point of failure, so a count of chains says nothing.

The factor two is charged exactly when adversarial text shares a reader’s context with honest evidence, which every partition of *relations* allows, since a forged copy of a relation is routed where the relation is. *Fact isolation*, one reader per fact and its document with the decision taken on the admitted facts, removes it. In the form $\omega \geq 2b + 1$ the limit is Dolev, Dwork, Waarts and Yung’s condition for reliable transmission over $2t + 1$ disjoint wires [25], with a witness as the wire; what is new is that the wires must be built from the evidence, that their number is a cut, and that the adversary can add wires as well as corrupt them.

End to end. We run fact isolation with a real LLM reader on MetaQA and Hetionet against four attacks, beside an undefended GraphRAG baseline (Section 7). No attack within the certified budget changed a report in 864 attacked runs; prompt injection never succeeds against fact isolation (0.0%, against 8.1% undefended); and the evaluation shows the limit’s other face, that on single-chain MetaQA the defense abstains and certifies almost nothing, while 72.0% of Hetionet’s treatment claims are certified end to end.

What partitioning relations costs. Every certified aggregation defense in the literature partitions relations without looking at the instance. Copying evidence into cells so that

every pattern lands somewhere intact is what a parallel database does when it schedules a join, and the minimum replication there is a linear program, the fractional edge packing number [1], [3]. Anchored at the question’s entity it prices the defense, and it is strictly below the arity d that independent-evidence certificates imply: on the 22 TPC-H queries the packing value is strictly smaller than the arity on **18 of the 20** multi-relation queries, so the certified budget is a median $3.2\times$ larger and up to $32\times$. That replication is still a cost the limit never pays: the packing number prices exactly one thing, refusing to look at the instance.

Contributions.

- 1) **The corroboration limit** (Section 5): every defense certifies at most $\lceil \tau/2 \rceil - 1$ forged relations, or $\tau - 1$ against an add-only adversary, and fact isolation attains both (Theorem 5.4); witness packing gives $\rho = 1$ and $\kappa = 1$ for every query (Theorem 5.2).
- 2) **The certificate is a max-flow** (Section 6): for multi-hop questions τ is a minimum cut (Corollary 6.1), answer selection is certified by its margin at every path length (Corollary 6.3), and the cut is the cheapest rewriting attack (Section 6.1).
- 3) **End-to-end evaluation** (Section 7): fact isolation with a real LLM reader on MetaQA and Hetionet, against shared-relation forgery, knowledge-evolution modification, a min-cut adaptive attacker and prompt injection, beside an undefended GraphRAG baseline under the same attacks.
- 4) **How much there is to certify** (Section 8): exact corroboration of 283,433 MetaQA answers, Hetionet questions and 1,510 drug–disease claims.
- 5) **The packing law** (Section 9, Appendix A): a content-oblivious partition pays amplification K^{1-1/τ_A^*} , optimal wherever the integral and fractional packings agree, and its majority certificate is capped by evidence redundancy. A one-command gate asserts every number in the paper against the committed data.

2. Related Work

Structural poisoning. Poisoning a learner by writing its training or retrieval corpus goes back to Biggio, Nelson and Laskov [36], and Carlini et al. [35] show it is practical at web scale. In the retrieval setting, adversarial passages crafted to be retrieved for unseen queries [34], PoisonedRAG [23] and BadRAG [22] inject at the corpus level, while indirect prompt injection [33] shows retrieved content can carry instructions as well as facts. Knowledge graphs are attackable in their own right: Zhang et al. [41] manipulate the plausibility of a target fact by adding or deleting edges. GraphRAG-specific poisoning is established: GragPoison [10] exploits *shared relations* to compromise many queries with one injection, and KEPo [4] poisons the knowledge-evolution step, anchoring forged facts to authentic ones and connecting poisoned communities across targets; LogicPoison [51] and targeted knowledge poisoning [52] corrupt the reasoning paths with a handful of edits. These operate at the layer our certificate prices.

Detection versus certification. A second line filters poison before it reaches the generator: post-retrieval filtering of adversarial passages [12], activation-space detection of poisoned responses [13], cluster filtering with model self-assessment [18], and information-flow control that keeps untrusted evidence away from the synthesising agent [6]; sparse document attention [54] forbids cross-attention between retrieved documents, an architectural cousin of the isolation rule of Section 5. The first defense built for GraphRAG itself, HoG-GRAG [16], decomposes a multi-hop question into ordered sub-queries, watches each hop for poisoning-induced inconsistencies and repairs the retrieved subgraph by pruning. All assume poison is identifiable before it influences the answer, which an adaptive adversary optimises against, and a forged relation anchored to authentic ones, which is what KEPo does, is built to pass a consistency check. A certificate bounds what *any* adversary within budget can do, detected or not. The two compose, since filtering shrinks the m_s a certificate must absorb.

Certified aggregation, and the gap they share. RobustRAG [21] isolates retrieved passages into *disjoint* groups so a corrupted passage spoils one group; ReliabilityRAG [14] does the same for RAG chunks; PRA-RAG [17] samples subsets of the retrieved passages and keeps the subset that is robust in embedding space, with a bound on the deviation the remaining poison can induce; Deep Partition Aggregation [9] and finite aggregation [19] certify poisoning budgets by partitioning, as does targeted partition aggregation for generation [53], bagging gives an intrinsic certificate of the same shape [38], and randomized smoothing [5] certifies ℓ_2 perturbations; the line begins with the approximate upper bounds of Steinhardt, Koh and Liang [37]. Certification on *graphs* exists but answers a different question: Bojchevski and Günnemann [39] and PGNNCert [40] certify a prediction against perturbations of the graph a model runs on, where we certify an answer against forged evidence retrieved into a prompt. Disjointness is available to all of them precisely because a passage means something alone: implicitly $\rho = 1$, and no replication factor appears in their analyses. Graph evidence removes that option, and pricing what it costs is this paper’s subject. The Lethal Dose Conjecture [20] posits partition aggregation is near-optimal among poisoning defenses; for relational evidence Theorem 5.4 settles the question exactly, and the unit it isolates is the fact, not the passage group or the relation. For adaptive attackers, Kaneko and Baldwin [8] bound attack efficiency by per-query leakage.

Reliable transmission and quorums. The form of our main theorem is classical: Dolev, Dwork, Waarts and Yung [25] give $2t + 1$ disjoint wires as the condition for reliable message transmission, the Byzantine Generals problem [44] fixes the arithmetic of surviving t liars, and Byzantine quorum systems [26] build availability from intersecting sets. There the wires are given by the system; here they must be extracted from an evidence hypergraph, their number is a transversal (Theorem 5.4) or a flow (Corollary 6.1), and the adversary adds evidence rather than corrupting channels. Max-flow has bounded adversarial influence before:

Advogato [46] and SybilLimit [47] cap what Sybil identities gain by the cut between honest and attacker regions of a trust graph, and SumUp [48] collects votes by flow so that bogus votes are bounded by attack edges. Data fusion detects copying between sources [49], the passive counterpart of our edge-disjointness. Those systems cut a social graph to bound identities; we cut the evidence of one answer to bound forged facts, and prove the cut is the limit for every defense. Concurrently, Pandey et al. [50] certify structured agent actions by the minimum hitting set of the control domains attesting each field, with an indistinguishability converse of the same kind; their evidence for a value is a set of attestations, ours a conjunction of relations, so their count is the single-fact case of τ , and the add/remove dichotomy, the cut and the packing law have no counterpart there.

Parallel query processing. The *shares*/HYPERCUBE algorithm [1], the parallel complexity of conjunctive queries [42] and the one-round load bound of Beame, Koutris and Suciu [3] identify τ^* as the exponent governing per-server load, with the skewed case treated separately [43]; the AGM bound [2] and worst-case-optimal joins [11] do the analogous job for the fractional edge *cover*. To our knowledge these have not been used as *security* parameters. The reading is not merely an application: the objective is adversarial influence rather than communication, the inner maximisation over poison sets has no counterpart in the load literature, and what we prove is a minimax, $\min_{\Pi} \max_{P: |P| \leq b} |\text{AffectedCells}_{\Pi}(P)|$. Remark T.1 shows where the two problems part company: co-location is strictly stronger than one-round evaluation, so the load bound does not transfer and we prove the lower bound directly.

3. Threat Model

System. A GraphRAG pipeline answers “does the relational pattern M hold around entity v ?” It retrieves a subgraph around v , partitions the N retrieved edges into K *cells*, submits each cell independently to a verdict function (an LLM or an exact subgraph checker), and aggregates the K binary verdicts. Instances include multi-hop question answering over a knowledge graph, provenance and citation chains, dependency reachability and transaction-cycle screening; the analysis fixes none of them.

Adversary. Suppress a true positive: make the pipeline report the pattern absent when it is present. Section 5 certifies the symmetric false-positive goal as well. An (m_d, m_s) -adversary may inject m_d forged **documents**, each entering one cell, and m_s forged **relations**, each entering every cell the partition assigns to that content. The second is *structural poisoning* and is the capability GragPoison and KEPo exercise. The adversary has unbounded computation, full knowledge of the algorithm, the graph and the motif, and may query the 1-bit aggregate verdict between injections. It cannot modify the partition code, the verdict function, or existing honest records: it can only add. Section 5 also prices a *neutralising* adversary that can remove or rewrite relations, as knowledge-evolution poisoning does.

Three budgets. Real attacks are mounted in *text*: forged documents enter crawlable sources and the pipeline’s own extractor turns them into relations. Writing B_{text} for injected documents, B_{edge} for distinct forged relations realised and B_{cell} for corrupted cell verdicts, Theorem E.1 orders them,

$$\begin{aligned} B_{\text{cell}} &\leq \rho_{\max} B_{\text{edge}} \\ &\leq \rho_{\max} c_{\text{edge}} B_{\text{text}}, \end{aligned}$$

with c_{edge} the extractor’s yield. We prove the left inequality; the right is a property of the extraction pipeline. Everything below is stated in $B_{\text{edge}} = m_s$. The certificate is with respect to the *realised* relation set: we do not assume extraction is honest, and an attacker who makes the extractor emit more relations per document moves the right inequality, not the left.

Scope. The theory needs one thing, that the answer is supported by a conjunction of facts joined on shared entities; M is not the user’s question but the *support pattern* grounding an answer. Verification queries and multi-hop entity QA are covered directly, bounded reachability as a union of path queries, counting one membership query per counted item, free-form generation per claim against a *declared* support class, and unbounded recursion only via truncation. Two properties make the coverage broad: $\tau_A^* \leq d$ always, so the LP never prices a pattern above the rate prior work assumes, and τ_A^* is monotone under adding atoms, so a query class is bounded by its largest member.

4. Preliminaries

Graph and motif. $G = (V, E)$ is directed with degree bound w and anchor $v \in V$. A *motif* M is a connected pattern with d edges. Read M as a conjunctive query $Q(\bar{x}) : -R_1(\bar{x}_1), \dots, R_d(\bar{x}_d)$ with *bound* variables $A = \{v\}$ and *free* variables $U = V(M) \setminus A$; N is the number of retrieved candidate edges per atom. Nothing before Section 8 uses the fact that atoms are binary. A *witness* is an assignment $\phi : U \rightarrow V$ making every atom present.

Defense. A ρ -replicated K -partition assigns each retrieved edge to ρ of K cells. Cell i fires ($h_i = 1$) iff the edges it holds contain a complete witness; the clean margin is

$$\mu = c_1 - \lfloor K/2 \rfloor,$$

with c_1 the number of firing cells, and B is the per-cell load, that is the verdict function’s context budget. We write $\bar{B}$ for the mean load, so $K\bar{B}$ is the total context the defense consumes.

Assumptions. Five, each named where it is needed.

A1 (cell-determinism)

Cell membership is a function of the edge content alone, so a forged edge is routed exactly as an honest one with the same content.

A2 (uniform replication)

Every edge occupies the same number of cells, ρ ; designs with unequal replication are covered by reading ρ as $\rho_{\max}$.

A3 (content-surjective adversary)

The white-box adversary can craft an edge whose cell-set is any reachable ρ -subset. Remark G.1 shows the deployed design does not admit this, which is what Section G recovers.

A4 (private hash)

Used only in Section I: the hash maps each content to an independent uniform ρ -subset, never revealed.

A5 (reached cells are flippable)

A cell whose evidence the adversary alters may return an arbitrary verdict. Used only for the *lower* bounds; the certificates need only the upper bound $|C_{\Pi}(e)| \leq \rho_{\max}$. Remark E.3 says when A5 is true.

5. The Corroboration Limit

Section 9 rests on one step: a cell fires only on a whole witness, so the same relation must be copied into many cells. That step is sound for designs that must place *every* retrieved relation somewhere. Dropping that requirement removes the copying, and with it the packing exponent.

Definition 5.1 (Witness hypergraph, and the two designs). Let H have the retrieved relations as vertices and the witnesses as hyperedges. Three numbers measure how much independent support it carries: the largest set of pairwise edge-disjoint witnesses, $\omega(H)$; the fractional witness packing number

$$\nu^*(H) = \max \sum_W y_W$$

subject to

$$\sum_{W \ni e} y_W \leq 1 \quad \forall e, \quad y \geq 0;$$

and the *transversal number* $\tau(H)$, the fewest relations meeting every witness. LP duality gives

$$\nu^*(H)/d \leq \omega(H) \leq \nu^*(H) \leq \tau(H).$$

The *witness-packing partition* at target t enumerates the witnesses of the *realized* instance, selects a *maximum* edge-disjoint family $W_1, \dots, W_\omega$, gives cell i exactly the tuples of W_i with their provenance documents, and reports $\hat{F} = 1$ iff at least $t + 1$ cells fire, abstaining otherwise. The target t is fixed by the deployment and never read off the instance: a threshold such as $\lceil \omega/2 \rceil$ of the realized packing is one the adversary can move, since on an instance with no witness at all a single forged relation completes one, makes $\omega = 1$, and clears any threshold that scales with ω .

Theorem 5.2 (Witness packing gives $\rho = 1$ for every query). *Under Definition 5.1, on any realized instance: (i) every cell fires, so $c_1 = \omega$; (ii) each forged relation lies in at most one selected witness, so it corrupts at most one cell and $\rho = 1$ for every query shape, with no hash and no share vector; (iii) a cell holding no forged tuple contains exactly its clean witness, so its verdict is unchanged; (iv) with $\Delta = m_d + m_s$ and target t , no instance without a witness is reported present unless $\Delta \geq t + 1$, and every instance with*

$\omega \geq 2t+1$ witnesses is reported present whenever $\Delta \leq t$; so the largest target certified in both directions on an instance with ω witnesses is

$$b_{\text{cert}} = \lceil \omega/2 \rceil - 1,$$

which is exactly the condition $\omega \geq 2t+1$, and this is exactly tight; and (v) no forged tuple reaches two cells, so verdict errors from shared forged evidence cannot be correlated: $\kappa = 1$ by construction and the $\sqrt{\kappa}$ fee disappears.

Where the factor two comes from. Theorem 5.2 charges a forged relation one cell and still halves ω , against an adversary that can only add. The reason is A1: a forged copy of an honest relation is routed into the cell of the honest witness that holds it, where its payload can silence that cell. Co-location hands an add-only adversary the power to *neutralise* honest evidence. Separating the two capabilities exposes the limit, and it is neither ω nor ν^* .

Definition 5.3 (Two adversaries, and fact isolation). Evidence is a set of tuples, each a relation with its provenance document. An *add-only* adversary of budget b inserts at most b tuples carrying documents of its choosing. A *neutralising* adversary may also remove honest relations, which is what modification does [4], with at most b insertions and removals in total. A defense reads documents only through *readers*: a reader shown only honest text is correct, and a reader shown adversarial text may return anything, which is A5. *Fact isolation* gives every tuple a cell of its own, asks the reader whether the document states the relation, admits a relation if any of its cells says yes, and decides on the admitted relation set $\hat{R}$ alone.

Theorem 5.4 (Corroboration is all you can certify). Let $\tau = \tau(H)$ on the clean retrieved evidence and let b_{cert} be the largest target certified in both directions. For every defense in the sense of Definition 5.3, whatever its cells, aggregator or compute:

- (i) against a neutralising adversary, $b_{\text{cert}} \leq \lceil \tau/2 \rceil - 1$;
- (ii) against an add-only adversary, $b_{\text{cert}} \leq \tau - 1$.

Fact isolation with the transversal rule, report present iff $\tau(H(\hat{R})) \geq t+1$, attains both. The fractional rule, $\nu^(H(\hat{R})) > t$, runs in polynomial time and certifies $\lceil \nu^*/2 \rceil - 1$ and $\lceil \nu^* \rceil - 1$ respectively. No bound depends on K , on a share vector or on τ_A^* .*

The converse is a meet in the middle and assumes nothing about the defense beyond its readers. Split a minimum transversal T into halves T_1, T_2 . Removing T_1 from the positive and adding T_2 to the negative $P \setminus T$ produce the same relations, and the adversary sets the readers of its own documents to agree with the honest ones, so the defense sees one instance and one of the two attacks succeeds. An add-only forger must supply all of T itself. This is the unique-decoding radius of a code whose distance is τ , and the $2t+1$ disjoint wires of Dolev et al. [25] are its special case.

The factor two is therefore neither the price of a fallible reader nor of defending two directions. It is the price of letting the adversary *remove* support, and a design grants

that power whenever a payload can share a reader's context with honest evidence. Under A1 a forged copy of r lands in every cell that holds r , so for a design whose cell layout is a function of relation content, such as Definition A.2, a co-located payload is a neutralisation and (i) applies to it. Fact isolation never shows a reader two documents, so an add-only adversary stays add-only and the factor two disappears. The rule for practice is short: **never let adversarial text share a reader with honest evidence.**

One chain, three designs. Since $\omega \leq \nu^* \leq \tau$, the designs line up: witness packing certifies $\lceil \omega/2 \rceil - 1$, the fractional rule $\lceil \nu^*/2 \rceil - 1$, and the transversal rule $\lceil \tau/2 \rceil - 1$, or $\tau - 1$ against an add-only adversary. Computing τ is a hitting-set problem, NP-hard in general, which is what the fractional rule buys back. For the questions of Section 6 the three coincide and equal a minimum cut. Where they differ, the gap is structural: on the anchored C_4 every witness uses two of the anchor's relations, so ω and ν^* are capped near $\deg(v)/2$ while τ is near $\deg(v) - 1$, and Section 8 measures the transversal rule certifying more than twice what witness packing does. Each budget is computed on the retrieved evidence, and what a content-oblivious partition pays instead is Corollary 9.1. Cells beyond what the evidence supports are worthless at any K , which is the final form of Corollary H.5 with no τ_A^* left in it.

6. The Certificate Is a Max-Flow

Enumerating witnesses is a join, and τ is a hitting set over them. For path-shaped support patterns, which is what multi-hop question answering and provenance checking are, neither is needed. Edge-disjoint witnesses are edge-disjoint paths and a transversal is an edge cut, so Menger's theorem [27] makes $\omega = \nu^* = \tau$ and turns the certificate into a flow, computable by any algorithm in the line Ford and Fulkerson [45] opened.

Corollary 6.1 (Certified robustness is the edge-connectivity). Fix a question entity v and a candidate answer a , and let ω_a be the maximum number of edge-disjoint support paths from v to a in the retrieved subgraph. Then ω_a is the minimum v - a edge cut, and the certified budget for that answer is

$$b_{\text{cert}}(a) = \lceil \omega_a/2 \rceil - 1$$

against a neutralising adversary, or under a design such as witness cells that co-locates payloads with honest evidence, and $\omega_a - 1$ against an add-only adversary under fact isolation. In particular, at two hops

$$\omega_a = |N(v) \cap N(a)|,$$

the number of distinct intermediate entities. When the support pattern is a typed path, a fixed sequence of relations as in every multi-hop benchmark question type, the witness graph is layered by hop and every v - a path has the same length, so the flow is exact at every length provided no relation type recurs at two hops that are both free of v and a . For untyped paths under a length bound $L \geq 5$ exactness is NP-hard [28], and a greedy family certifies within a factor d (Remark 6.2).

The certificate is therefore defined *per answer*, which is what a question-answering system needs, and it costs one max-flow rather than an enumeration. It also says plainly what a deployment can promise: *this answer is supported by ω_a independent chains, and no $\lceil \omega_a/2 \rceil - 1$ forged relations can change it*. The rule and the certificate are one object: with the deployment’s target t , report a iff the flow over the admitted relations is at least $t + 1$. When ω_a falls below what t needs, the system should abstain rather than answer, which turns the ceiling into a deployable rule rather than a limitation.

Remark 6.2 (The selection must be maximum). Any edge-disjoint family lower-bounds ω , but as a *selection rule* an approximate packing is not adversary-proof. A forged witness that a greedy rule picks first holds one forged tuple and $d - 1$ honest ones, so it blocks up to $d - 1$ honest witnesses and corrupts its own cell: one forged relation displaces up to d cells, $d - 1$ of them honest, and the $\rho = 1$ of Theorem 5.2(ii) becomes $\rho = d$. A maximum packing cannot shrink when tuples are added, which is the step the proof of (iv) uses. Path-shaped patterns get a maximum packing from the flow; for general patterns it is a set-packing integer program over the retrieved witnesses, which are few.

Corollary 6.3 (Answer selection: the budget is the margin). *Let the system report the candidate answer of largest corroboration τ_a over the admitted relations, abstaining on a tie, and let $\Delta = \tau_a - \max_{a' \neq a} \tau_{a'}$ on clean evidence. Under fact isolation no adversary within*

$$b_{\text{cert}} = \Delta - 1$$

insertions and removals changes the report, at every path length. A witness-cell design with co-located payloads guarantees $\Delta - 1$ at two hops but only $\lceil \Delta/2 \rceil - 1$ at three or more.

Proof in the appendix. The margin is not halved because a comparison has no fixed threshold to defend from both sides: one insertion raises a competitor by at most one, one removal lowers the leader by at most one, and neither does both. Witness cells lose this at three hops, where one forged relation can complete a witness of a and of a' at once. This is also where counting fails and corroboration does not. A hundred witnesses that all run through one relation are one piece of evidence with a hundred spellings: their edge-disjoint count is 1, a single forged relation at the shared position flips the answer, and the certificate correctly reports nothing. Numerosity is also manufacturable. For patterns of three hops or more, one forged relation at a bottleneck turns every pre-existing path through that position into a fresh witness, which is exactly the shared-relation amplification the attack literature exploits [10]. Raising a cut costs the adversary one forged relation per unit; raising a raw count can cost it one in total. When the margin is zero, as it is for the median answer measured in FLOW-CHECK, the honest report is abstention rather than either answer.

TABLE 1. WHICH CEILING APPLIES. THE FACTOR TWO IS CHARGED EXACTLY WHEN THE ADVERSARY CAN REMOVE SUPPORT, BY MODIFYING RELATIONS OR BY A PAYLOAD THAT SHARES A READER WITH HONEST EVIDENCE. FOR A TYPED MULTI-HOP QUESTION $\omega = \nu^* = \tau$ IS ONE MINIMUM CUT.

setting	ceiling	attained by
add-only, payload isolated	$\tau - 1$	fact isolation
add-only, co-located	$\lceil \tau/2 \rceil - 1$	witness cells: $\lceil \omega/2 \rceil - 1$
neutralising (modify)	$\lceil \tau/2 \rceil - 1$	fact isolation
polynomial time	ν^* for τ	fractional rule

6.1. The same LP, read from the other side, prices the attack

The packing LP has a dual. By LP duality

$$\nu^*(H) = \tau^*(H) = \min \sum_e x_e$$

subject to

$$\sum_{e \in W} x_e \geq 1 \quad \forall W, \quad x \geq 0,$$

the minimum *fractional* transversal, a lower bound on $\tau(H)$. An adversary who may only *add* relations is optimally served by injecting $\lceil \omega/2 \rceil$ pairwise edge-disjoint fresh witnesses against witness packing, the tightness construction of Theorem 5.2(iv), and by injecting τ against fact isolation. An adversary who may *modify* existing relations, which is exactly what knowledge-evolution poisoning does [4], must rewrite a transversal, at cost exactly $\tau(H)$. For a multi-hop answer that is a minimum edge cut, found by the max-flow that computes the certificate, so the cheapest structural poisoning of a multi-hop answer is computable in polynomial time and published structural attacks, which are heuristic, can be measured against it. For general patterns the minimum transversal is a hitting set; rounding the LP gives a logarithmic approximation. Both sides are priced by one LP and its integral counterpart, and the procedure applies to undefended pipelines as much as to defended ones.

What the designs cost. Witness packing leaves evidence outside the selected packing unread, which lowers recall and leaves the certificate unchanged, and each cell shows the reader d facts rather than a neighbourhood. Fact isolation reads every retrieved tuple once and shows each reader one document, so cross-fact context is gone by construction and the reasoning moves into τ ; it is one reader call per tuple. Witness enumeration is a join, cheap for small support patterns and bounded by the AGM exponent in general [2]. Soundness under adaptive injection does not depend on telling forged tuples from honest ones. Certifying the full $\tau - 1$ sets the threshold at the evidence’s own corroboration, so one honest relation lost to retrieval noise flips it; a deployment trades the two by its choice of t .

Two supporting facts are in Appendix W: randomising the selection cannot improve Theorem 5.2, and for the small ω that real evidence yields the exact binomial tail certifies where the Chernoff form of Theorem H.3 is empty.

TABLE 2. END TO END, ATTACK SUCCESS IN PER CENT. METAQA: THE DEFENDED AND UNDEFENDED ANSWERS FLIPPED TO THE ATTACKER’S TARGET, AND THE DEFENDED REPORT CHANGED AT ALL (MOST CHANGES ARE TO ABSTENTION). HETIONET: THE DEFENDED DECISION FLIPPED. EVERY CHANGE SPENT MORE OPERATIONS THAN THE CLEAN CERTIFICATE.

attack	<i>b</i>	MetaQA		Hetionet	
		def. flip	undef. flip	def. changed	def. flip
shared-relation	1	0.0	0.0	71.6	20.8
	3	14.9	1.6	87.8	83.3
modification	1	77.8	84.9	96.8	8.0
	3	84.1	86.8	100.0	36.0
min-cut adaptive	1	0.0	0.0	78.4	14.3
	3	14.9	1.6	86.5	59.2
prompt injection	1	0.0	8.1	0.0	0.0
	3	0.0	8.1	0.0	0.0

7. End-to-End Evaluation

Theorem 5.4 assumes one thing about the world, that a reader shown honest text alone is correct, and promises one thing, that no attack within the certified budget changes the report. `e2e_factiso.py` tests both with a real model.

Setup. The reader is `qwen-122b`, an open-weight model served by `vLLM` with reasoning disabled, asked of each (fact, document) pair in its own call whether the document states the fact. On MetaQA the document is the movie’s WikiMovies article [55]; we take 200 two-hop test questions, at most three per template, whose typed two-hop neighbourhood has at most 60 facts (a mean of 6.4), and report the answer of largest max-flow, abstaining on a tie, with budget $\Delta - 1$ (Corollary 6.3). On Hetionet we check 100 treatment claims and 50 non-claims under the seven-metapath class of CLAIM-CHECK, accepting at $\tau \geq 2$ ($t = 1$); its documents are template sentences, which makes the reader’s job easy, so MetaQA is the main dataset. The undefended baseline puts the same retrieved facts and documents into one prompt and asks the question. Each attack is run at budgets 1 and 3: *shared-relation* forgery in the style of GragPoison, attaching forged relations, each with an LLM-written article, to the most-shared intermediates to promote a fresh answer; *modification* in the style of KEPo, rewriting relations that support the clean answer, with an LLM-written “updated” article (one removal and one insertion each); a *min-cut adaptive* attacker that reads the certificate, targets the strongest competitor and, against a claim, deletes a minimum transversal; and *prompt injection*, attaching “answer NO” documents to the relations in the clean answer’s cut and adding relations whose documents state nothing but “answer YES”.

The guarantee holds. Over 864 attacked runs the defended report changed 436 times, and every one of them spent more operations than its clean certificate allowed: no attack within the budget succeeded, exactly as Theorem 5.4 predicts.

Prompt injection is inert. Against fact isolation injection succeeds 0.0% of the time, against 8.1% undefended.

The reader answered yes to 0.0% of the injected documents, and an “answer NO” document attached to an honest relation sits in a cell of its own, so it cannot remove the relation however the reader responds. This is the factor two of Theorem 5.4 made concrete: an injection that shares the reader’s context can remove support, and one that does not, cannot.

What the attacks buy. Forged documents that state their relation are admitted (85.1%), as the theorem allows: they are insertions, and they are charged as such. On MetaQA the clean margin is the problem: the median Δ is 0, only 3 of 200 questions carry a certificate of one relation, and three forged chains promote a fresh answer in 14.9% of questions, where the undefended model, which reads the documents, is fooled in 1.6%. Most defended changes go to abstention (240 of 364), not to the attacker’s answer. Modification removes support, the neutralising adversary, and flips both systems alike. On Hetionet the claims are corroborated, and forgery needs budget 3 to accept a non-claim in 83.3% of cases, each time beyond the certificate.

Clean accuracy and certification. On MetaQA fact isolation answers 37.0% of questions correctly and abstains on 63.0%; with a perfect reader it would answer 50.0%, the rest being ties in the evidence itself, and it certifies 1.5% at $t = 1$. The undefended model answers 74.5%, with no certificate. On Hetionet fact isolation is right on 83.3% of claims and certifies 72.0% of true claims at $t = 1$, 53.0% at $t = 2$ and 34.0% at $t = 3$, matching CLAIM-CHECK; the undefended model accepts only 3.0% of true claims, judging the support indirect, so its Hetionet attack rows carry no information and are omitted.

The reader’s assumption. On honest MetaQA text the reader says no to 11.9% of facts whose object the article names, and to 0.0% of Hetionet’s template sentences. The rate is 1.1% for *directed by* and 2.6% for *starred*, while *written by* (37.6%) and *language* (66.7%) are mostly the knowledge base claiming what the article does not say, a director named as writer, a language never stated. A false negative lowers τ ; it never admits a forgery, so it costs accuracy and certified coverage, not soundness.

Cost. A query costs one reader call per fact, 7.4 on MetaQA and 11.4 on Hetionet; at 0.50 s per call that is 0.5 s and 1.0 s per query with eight calls in parallel. The whole evaluation took 5,396 calls.

8. Real Workloads

Section 7 runs the certificate; this section asks how much there is to certify. Every quantity is exact: typed paths make the witness graph layered, so $\omega_a = \tau$ is a max-flow, and union support classes are solved by integer program.

METAQA and HETIONET: real workloads. A power grid is not where GraphRAG runs, so `kg_omega.py` measures ω_a where it does: on MetaQA, the standard multi-hop KGQA benchmark over the WikiMovies graph (134,741 triples), scoring every gold answer of every two- and three-hop test question, 92,323 and 191,110 answers, with each template’s relation path inferred from the graph as the typed

path whose endpoints best match the gold answers (choosing the path with the gold answers in view can only favour well-supported paths, so it biases ω_a upward, against our finding); and on Hetionet (47,031 nodes, 2.25M typed edges), the Medical domain of the GragPoison benchmark [10], asking four metapath questions, among them the introduction’s own *which compound binds a gene this gene regulates*, at 300 anchors each. The flow is exact because the paths are typed (Corollary 6.1), and the disjointness of the extracted witnesses is re-checked on every answer. Table 3 is Figure 2 drawn larger: the median answer rests on one chain everywhere; 96.4% of MetaQA’s two-hop answers and 92.5% of its three-hop answers are single-chain, and fewer than 1.4% of either carry the three independent chains that certify one forged relation; on Hetionet the single-chain share runs from 63.5% to 93.7% by question. Well-corroborated answers exist, up to 22 chains on MetaQA and 16 on Hetionet, and they are a few per cent of the whole. (2,261 three-hop answers (1.2%) have no chain that avoids the question entity or the answer, which MetaQA’s set-following generation admits, and sit in the $\omega_a = 0$ bin.) The paths are typed, so $\omega_a = \tau$ and, read with Theorem 5.4, this is how much of a single-template KGQA workload *any* defense can certify as retrieved: almost none of it.

CLAIM-CHECK: the support class decides. A single template is the narrowest support class there is. `kg_claims.py` asks the question claim verification asks: does Hetionet corroborate “compound c treats disease d ” with the direct edge removed, under a declared class of seven metapaths (shared target genes, shared up- and down-regulated genes, similar compounds, related diseases, and a pharmacologic class), over all 755 treatment edges and 755 random compound–disease pairs with no such edge. Here $\omega = \nu^* = \tau$ on every pair. The last two rows of Table 3 show the result: the median true claim rests on 5 independent chains and 72.3% certify at least one forged relation, against 11.7% of non-claims, so the same threshold that certifies is also a classifier. (Some non-claims may be unlisted treatments.) The abstention rule, a declared support class wide enough to corroborate, and a retriever that seeks corroboration are the operative advice; a larger K is not.

FRACTIONAL-GAIN and FLOW-CHECK. `menger_check.py` computes ω , ν^* and τ exactly at the PEGASE anchors of Appendix J, by set-packing ILP, LP and hitting-set ILP, and the chain $\omega \leq \nu^* \leq \tau$ holds at every anchor. On the 2-path the three are equal at every anchor. On the anchored C_4 the ratio ν^*/ω reaches a maximum of 1.50 and is strictly above 1 at 7 of 12 anchors, but the large gap is the integral one: τ exceeds ν^* by up to $1.85\times$, with median $\tau = 11$ against $\omega = 6$. The transversal rule of Theorem 5.4 therefore certifies a median of 5 forged relations on the C_4 where witness packing certifies 2, from the same evidence. For the flow reading, we compare the number of edge-disjoint witnesses of each two-hop answer with its common-neighbour count over 464 answers: they agree in 464 of 464, so Corollary 6.1 is exact on this instance and the certificate needs no enumeration.

The distribution (Figure 2) is bimodal rather than merely

TABLE 3. METAQA, HETIONET AND CLAIM-CHECK. INDEPENDENT SUPPORT $\omega_a = \tau$ PER GOLD ANSWER OR CLAIM, EXACT BY MAX-FLOW OR ILP. “= 1” IS THE SINGLE-CHAIN SHARE; “ ≥ 3 ” IS THE SHARE THAT CERTIFIES AT LEAST ONE FORGED RELATION; “ ≥ 7 ” CERTIFIES THREE.

question	answers	= 1	≥ 3	≥ 7	max
<i>MetaQA</i> (WikiMovies)					
two-hop (210 templates)	92,323	96.4%	0.9%	0.1%	22
three-hop (150 templates)	191,110	92.5%	1.3%	0.0%	9
<i>Hetionet</i> (C compound, G gene, D disease)					
C–G–D	5,841	66.0%	15.2%	1.7%	15
G–G–C	10,656	93.7%	1.0%	0.0%	5
C–C–D	646	63.5%	20.0%	3.3%	13
C–D–G–C	186,015	84.5%	4.5%	0.2%	16
<i>Hetionet claims</i> , c treats d , seven metapaths					
true claims	755	11.3%	72.3%	35.2%	27
non-claims	755	17.2%	11.7%	2.5%	24

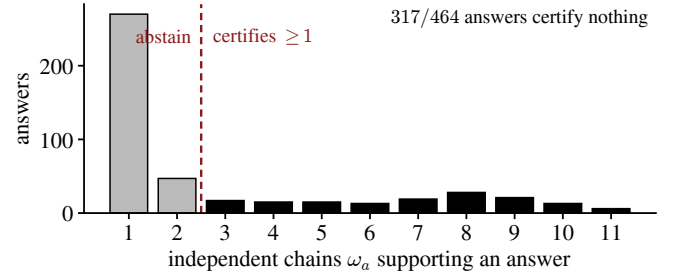


Figure 2. FLOW-CHECK: how much independent support the retrieved evidence actually carries, over all 464 two-hop answers on PEGASE. The distribution is bimodal — 270 answers rest on a single chain, while 87 carry seven or more. Everything left of the dashed line certifies nothing and is where Corollary 6.1 says to abstain.

thin. The median answer has $\omega_a = 1$ and the best has 11; 317 of 464 answers certify nothing at all, while 87 carry seven independent chains or more. The evidence a deployment needs is present for a minority of answers and absent for most, which is a measurement of the corpus, not a limitation of the defense.

9. What Partitioning Relations Costs: the Packing Law

Every certified aggregation defense in the literature partitions *relations*, without looking at the instance, and the question this section answers is what that costs. A cell can check a pattern only if it holds every relation of some witness, so a content-oblivious partition must copy each relation into ρ cells, and by A1 a forged copy lands in all of them: one forged relation corrupts $\rho_{\max}$ cells (Theorem E.1), which is the amplification behind the majority certificate $\mu > m_d + \rho m_s$ of Theorem E.2. Copying evidence so that every pattern lands somewhere intact is what a parallel database does when it schedules a join across servers, and the minimum replication there is a linear program on the query, the fractional edge packing number. Anchored at the

question’s entity, the same LP τ_A^* (Definition A.1) prices the defense:

- the anchored HYPERCUBE detects every witness with $\rho_{\max} = \Theta(K^{1-1/\tau_A^*})$, and no content-oblivious design that detects deterministically does better when the integral packing equals τ_A^* , which it does on all 35 query families and benchmark queries we evaluate (Theorem A.3);
- $\tau_A^* \leq d$, with equality only when the free variables form a matching, so the $K^{1-1/d}$ rate that independent-evidence certificates imply is not a floor;
- the margin μ depends on the same share vector, so the replication-optimal design often lights a handful of cells and the majority certificate is vacuous (Proposition H.1); thresholding on the firing count caps the budget at half of what the clean evidence lights (Corollary H.4).

Appendices E–J carry the full analysis and its measurements. What it amounts to is one comparison with Section 5:

Corollary 9.1 (The packing exponent is the price of obliviousness). *A content-oblivious design pays $\rho_{\max} = \Theta(K^{1-1/\tau_A^*})$ and needs $K^* = c_1^{\tau_A^*}$ cells to reach a budget of $\Theta(c_1)$; the witness partition reaches it with $\rho = 1$ and $K = \omega$ cells. The separation is polynomial in the replication rate and exponential in the cell count, and it does not contradict Theorem A.3(ii), which bounds content-oblivious designs only.*

TPC-H: a real query workload. We use the 22 TPC-H queries [15] over the published foreign-key schema, encoded table-by-table in the artifact so the encoding can be checked line by line rather than taken on trust. $\tau^* < d$ on 18 of the 20 multi-relation queries (Table 4) with share-LP duality gap at most 5.6×10^{-17} ; the median certified-budget gain is $3.2\times$ and the maximum $32\times$ at $K = 1024$; every two-table foreign-key join has $\tau^* = 1$, so the benchmark’s most common shape needs no replication at all where the arity reading demands $\sqrt{K} = 32$; the arity reading is wrong on 11 of the 20 queries as a cell-sizing rule; and anchoring can erase the gain entirely (Q17, Q19), which is the honest boundary of the result. These ratios are exponent ratios $K^{1/\tau_A^* - 1/d}$; the finite- K figures an oblivious design realises carry the $\sigma \leq d$ slot constant of Section 9.

10. Discussion and Limitations

Falsifications. Twelve hypotheses are falsified in print, all of them ours, and Table 5 lists them beside what survived. A framework that cannot falsify its own predictions is not being tested.

Beyond graphs. No theorem in this paper uses graph structure: the proofs assume evidence pieces, a conclusion that is a conjunction over *shared variables*, a content-determined partition and aggregation by vote. The artifact reproduces the calculus on an off-graph question over ternary relations, obtaining $\tau_A^* = 2 < d = 3$, $P_{\det} = 1$ and certificate tightness to the integer; the same reading applies to provenance chains, multi-source fusion (a star, the one shape where the arity reading is right) and SQL workloads, where the optimizer already computes the exponent.

TABLE 4. TPC-H. THE ARITY READING OVER-PRICES 18 OF THE 20 MULTI-RELATION TPC-H QUERIES. BUDGET RATIO IS THE CERTIFIED GAIN $K^{1/\tau_A^* - 1/d}$ AT $K = 1024$; **BOLD** MARKS A QUERY WHERE THE ARITY READING ALSO GIVES THE WRONG CELL-SIZING ADVICE.

Query	d	τ^*	τ_A^*	budget ratio
Q2	5	3	3	2.52×
Q3	3	2	2	3.17×
Q4	2	1	1	32.0×
Q5	6	3	3	3.17×
Q7	5	2	3	2.52×
Q8	7	4	4	2.10×
Q9	6	3	3	3.17×
Q10	4	2	2	5.66×
Q11	3	2	2	3.17×
Q12–Q15, Q22	2	1	1	32.0×
Q16	3	2	2	3.17×
Q17, Q19	2	1	2	1.00×
Q18	3	2	2	3.17×
Q20	5	2	3	2.52×
Q21	4	2	3	1.78×

What the ceiling asks of retrieval. Since the budget is τ of what retrieval returned, the objective that maximises certifiable robustness is not relevance. A retriever returning a hundred witnesses that share one relation certifies nothing; one returning ten edge-disjoint witnesses certifies four. Maximising the cut under a token budget is a min-cost flow for path-shaped patterns, so it is as cheap as the ranking it would replace, and the median $\omega_a = 1$ of FLOW-CHECK, METAQA and HETIONET says the objective is not currently being served. We do not evaluate such a retriever here and regard it as the natural next system.

What is not covered. A malicious relation extractor, malicious model weights, coordinated hallucination by cells holding *clean* evidence, an attacker who rewrites the query parser, and one who alters the partition computation. Those five are outside the model and we claim nothing about them. Within it, the outstanding gaps are four. The end-to-end evaluation uses one reader model, reimplements GragPoison and KEPo from their descriptions rather than running their code, and gives Hetionet template documents; its reader is wrong on honest text often enough (11.9% on MetaQA) that the assumption of Theorem 5.4 holds only approximately, which costs coverage but not soundness, since a false negative never admits a forgery. We do not measure κ for the HyperCube design, which fact isolation makes moot. Computing τ is NP-hard for general patterns, so off paths a deployment certifies the polynomial $\lceil \nu^*/2 \rceil - 1$, losing the integrality gap measured on the C_4 . And we do not close the gap between the integral packing ν used in Theorem A.3(ii) and the fractional τ_A^* achieved by (i); they agree on every family we evaluate.

Ethics. Read from the adversarial side, the tightness result yields an optimal attack against the defense this paper proposes. The attack assumes white-box knowledge of the partition, which we recommend keeping private in deployment. The experiments use only synthetic graphs, a

public benchmark workload, a public infrastructure topology and two public knowledge graphs; they do not involve live systems, proprietary or personal data, or human subjects. The released artifact is intended to support defensive evaluation and reproducibility. We therefore judge disclosure to provide net defensive value, while recognising that the attack construction could be misused against deployments that implement the evaluated partitioning strategy.

11. Conclusion

An isolate-and-aggregate defense is a one-round parallel query plan, so certified robustness against structural poisoning is a fractional edge packing LP and the arity rate implied by independent-evidence certificates is not a floor. That is the first half, and it prices only the denominator of the certified budget.

The numerator is evidence redundancy, and its limit holds for every defense: no design certifies more than $\lceil \tau/2 \rceil - 1$ forged relations against an adversary that can remove support, or $\tau - 1$ against one that can only add, where τ is the fewest relations meeting every witness, and isolating facts so that no payload shares a reader with honest evidence attains both. For a multi-hop question τ is a minimum cut, so the certificate and the cheapest rewriting attack are one max-flow. The packing number of the query then prices exactly one thing: refusing to look at the instance.

To a practitioner: compute the cut your retrieval actually returns, because that number is your budget; never let adversarial text share a reader with honest evidence; abstain below the target; return the verdict rather than the scores; and if you must partition relations, compute τ_A^* , partition by shares, and rebalance hot buckets instead of pinning them.

Partitioning can distribute evidence, but it cannot manufacture independent corroboration: the defender and the query optimizer have been solving the same problem all along, and what neither can buy is corroboration that was never retrieved.

The appendix holds the proofs of the main results; reviewers are not required to read it.

Appendix A. The Packing Law in Full

Everything above is monotone in ρ , so one question remains: how small can the replication factor be, subject to detecting the query at all?

The natural guess, and why it is wrong. A cell fires only if it holds a whole witness, so the design must guarantee that some cell contains all d tuples of any witness. Reading that as “every d -subset of the N retrieved tuples must be co-located” yields $\rho \geq K^{1-1/d}$, which is the natural extrapolation of independent-evidence certificates [9], [19]. It is correct *only* when any d tuples could constitute evidence, that is when the query is a matching. Relational evidence never is: the d tuples must agree on the variables they share, so the design must cover the $O(N^{\tau^*})$ actual witnesses, not

the $\binom{N}{d}$ subsets. For a k -cycle that is $N^{k/2}$ against N^k , and the counting is out by a square.

Definition A.1 (Anchored fractional edge packing). For a conjunctive query Q with atoms $\mathcal{A}(Q)$, bound variables A and free variables U ,

$$\tau_A^*(Q) = \max \sum_{R \in \mathcal{A}(Q)} x_R$$

subject to

$$\sum_{R \ni u} x_R \leq 1 \quad \forall u \in U, \quad 0 \leq x \leq 1.$$

Bound variables carry no constraint: their coordinates are pinned, not searched. An atom whose join variables are all bound still ranges over its rows, so it keeps a private free variable, its row identifier; that is the box $x_R \leq 1$, which binds only for such atoms.

Definition A.1 has two immediate properties. $\tau_A^*(Q) \leq d$ always, with equality *iff* every free variable occurs in exactly one atom, i.e. *iff* Q restricted to its free variables is a matching; and τ_A^* is monotone under adding atoms. Examples:

$$\tau_v^*(C_k) = \lceil k/2 \rceil,$$

$$\tau_v^*(K_{1,d}) = d \quad (\text{the star, the extreme case}),$$

$$\tau_v^*(b\text{-way scatter-gather}) = 2,$$

the last for every b , while that shape has $d = 2b + 1$ atoms. The exponent a defense pays is $1 - 1/\tau_A^*$ where the arity reading charges $1 - 1/d$: 0 against 0.50 for the 2-path, 0.50 against 0.75 for the C_4 , 0.50 against 0.86 for scatter-gather with $b = 3$, and equal only for the star, whose free variables form a matching.

Definition A.2 (Anchored HyperCube). Assign share exponents $a_u \geq 0$ with $\sum_{u \in U} a_u = 1$ and set $p_u = K^{a_u}$, so $\prod_u p_u = K$. Draw independent hashes h_u to $[p_u]$. A cell is a point $c \in \prod_u [p_u]$, and a retrieved tuple t is placed in cell c *iff* for some atom R that t could fill, $c_u = h_u(t[u])$ for every free variable $u \in \bar{x}_R$. This is the HYPERCUBE/shares construction of Afrati–Ullman [1] and Beame–Koutris–Suciu [3], with the anchors pinned.

Theorem A.3 (Packing law). *Fix a conjunctive query Q with bound variables A and K cells.*

- (i) (Achievability.) *The anchored HyperCube with shares solving $\max_a \min_R \sum_{u \in \bar{x}_R \cap U} a_u$ achieves deterministic detection, $P_{\det} = 1$, with*

$$\rho_{\max}(K) = \Theta(K^{1-1/\tau_A^*(Q)}),$$

so the amplification exponent is $\alpha^(Q, A) = 1 - 1/\tau_A^*(Q)$. The share LP value is exactly $1/\tau_A^*(Q)$.*

- (ii) (Optimality.) *Write $S_R = \text{vars}(R) \cap U$ and let $\nu(Q, A)$ be the largest set of atoms with pairwise disjoint supports, an integral packing, so $\nu \leq \tau_A^*$. Any content-oblivious design placing each tuple in at most ρ cells, leaving no cell with more than β times the mean number*

of tuples of any one atom, and detecting deterministically on every instance, obeys

$$\rho \geq \frac{K^{1-1/\nu(Q,A)}}{\beta}.$$

On every family we evaluate $\nu = \tau_A^*$, so (i) is optimal to the exponent.

(iii) (The arity bound is not a floor.) For every Q that is not a matching on its free variables, $\tau_A^* < d$ and the design of (i) satisfies $\rho_{\max} = o(K^{1-1/d})$ while keeping $P_{\det} = 1$.

Proof in the appendix, which also explains, in Remark T.1, why the one-round load bound does not transfer.

Appendix B.

Proof of Theorems 5.2 and 5.4

(i) holds because each cell contains a complete witness by construction. For (ii), the selected witnesses are pairwise edge-disjoint, so a tuple belongs to at most one of them and hence to at most one cell; a forged document attaches to one cell. (iii) is immediate from (ii). For (iv), fix the target t . Forgery: on an instance with no witness, every selected witness of the realized instance contains a forged tuple and the selected witnesses are pairwise edge-disjoint, so lighting $t+1$ cells consumes $t+1$ distinct forged tuples. Suppression: on an instance with $\omega \geq 2t+1$ witnesses, adding tuples only adds hyperedges, so a maximum packing of the realized instance has $\omega' \geq \omega$ members, at most Δ of which contain a forged item; the remaining $\omega' - \Delta \geq \omega - t \geq t+1$ cells hold exactly their clean witness and fire. Tightness: $t+1 = \lceil \omega/2 \rceil$ pairwise edge-disjoint forged tuples completing $t+1$ pairwise edge-disjoint witnesses are selected into $t+1$ distinct cells and forge a negative; when $\omega = 2t+1$ the same number of forged tuples, one placed in each of $t+1$ selected witnesses, silences them and suppresses a positive. (v) restates (ii).

Theorem 5.4, converse. Fix a defense, a positive P and a minimum transversal T of $H(P)$, $|T| = \tau$; then $N = P \setminus T$ has no witness. (i) Split $T = T_1 \sqcup T_2$ with $|T_1| = \lfloor \tau/2 \rfloor$ and $|T_2| = \lceil \tau/2 \rceil$, and suppose the defense certifies $t \geq \lceil \tau/2 \rceil$. The suppressor removes T_1 from P ; the forger adds tuples carrying the relations of T_2 to N , with documents of its own whose readers it sets to the verdicts the honest documents of T_2 receive. Both realised instances have the relation set $P \setminus T_1$, every honest document is shown to a correct reader in both, and the forger's documents read exactly as the honest ones, so the defense receives the same input and returns the same report. Both attacks are within budget, so one of them succeeds: a contradiction. (ii) If $t \geq \tau$, the forger adds all of T to N in the same way, and the defense sees P itself, which it must report present.

Theorem 5.4, achievability. Under fact isolation every honest tuple sits in a cell of its own, so its reader is correct and every honest relation that is not removed is admitted; an adversarial tuple can at most admit its own relation. Write D and A for the removed and the adversarially admitted relations, $|D| + |A| \leq t$. Removing a relation lowers τ by at

most one, since a transversal of $H \setminus e$ plus e is one of H , and adding relations only adds witnesses, so $\tau(H(\hat{R})) \geq \tau(P) - |D|$. On a negative every witness of $H(\hat{R})$ contains a relation of A , so A is a transversal and $\tau(H(\hat{R})) \leq |A|$. The rule $\tau(H(\hat{R})) \geq t+1$ therefore rejects every attacked negative, and accepts every attacked positive whenever $\tau(P) - |D| \geq t+1$: with $|D| = 0$ that is $\tau \geq t+1$, and in general $\tau \geq 2t+1$. For the fractional rule the same two steps hold with ν^* : deleting vertex e removes witnesses of total weight at most one, so ν^* drops by at most one, and on a negative $\nu^*(H(\hat{R})) = \tau^*(H(\hat{R})) \leq |A|$. $\square$

Appendix C.

Proof of Corollary 6.3

Under fact isolation the admitted set changes by insertions and removals only. An insertion adds witnesses, so it lowers no $\tau_{a'}$ and, as above, raises each by at most one; a removal raises none and lowers each by at most one. Either way the margin of a over each competitor falls by at most one per operation, and the report changes only when some margin reaches zero, which takes Δ operations.

For witness cells with co-located payloads, the packing of each answer is selected separately. At three hops let (v, x^*) , (y, a) and (y, a') be honest and forge (x^*, y) . It completes a witness of a that shares (y, a) with an honest one, which a maximum packing may select in its place, and whose cell the payload then silences, and it completes a witness of a' , whose new cell it lights: one relation, margin two. At two hops the witnesses of different answers share only a relation (v, x) at the anchor. A copy of it can silence the cell of a through x , but the cell of a' through x already fires; a forged (x, a') lies in no witness of a . So each forged relation moves the margin by one at two hops and by at most two in general, giving $\Delta - 1$ and $\lceil \Delta/2 \rceil - 1$. $\square$

Appendix D.

Proof of Theorem A.3

(i) **Detection.** Let ϕ be a witness. Every atom R is realized by a tuple t_R with $t_R[u] = \phi(u)$ for each free $u \in \bar{x}_R$, which by construction is placed in every cell agreeing with ϕ on $\bar{x}_R \cap U$, in particular in $c^\phi := (h_u(\phi(u)))_{u \in U}$. So all d tuples lie in c^ϕ and $P_{\det} = 1$ for every hash draw and every instance.

(i) **Replication.** A tuple filling R is replicated over the coordinates of $U \setminus \bar{x}_R$, i.e. $K / \prod_{u \in \bar{x}_R \cap U} p_u$ cells, so with $p_u = K^{a_u}$,

$$\rho_{\max} = K^{1-\gamma},$$

$$\gamma := \max_{a \in \Delta_U} \min_R \sum_{u \in \bar{x}_R \cap U} a_u.$$

Here γ denotes the value of the share LP, which is distinct from the integral packing number $\nu(Q, A)$ appearing in

part (ii). It is the value of the zero-sum game with payoff $M_{R,u} = \mathbf{1}[u \in \bar{x}_R]$. By von Neumann minimax,

$$\gamma = \min_{x \in \Delta_A} \max_u \sum_R x_R M_{R,u};$$

writing a feasible packing as $x = X/\lambda$ with $X \in \Delta_A$, feasibility is exactly $\max_u (M^\top X)_u \leq \lambda$, so

$$\tau_A^* = \max_X (\max_u (M^\top X)_u)^{-1} = 1/\gamma.$$

(ii) Optimality. Exhibit one instance: the complete instance over a domain of size D , where atom R has $N_R = D^{|S_R|}$ candidate tuples and every value of every variable occurs equally often, so the instance is skew-free. Let $P = \{R_1, \dots, R_\nu\}$ be a maximum integral packing and $T_i(c) = \{t : c \in f_{R_i}(t)\}$ the tuples of R_i routed to cell c . Since $\sum_c |T_i(c)| = \sum_t |f_{R_i}(t)| \leq \rho N_i$, balance gives $|T_i(c)| \leq \beta \rho N_i / K$ for every c . The supports $S_{R_1}, \dots, S_{R_\nu}$ are pairwise disjoint, so as ϕ ranges over D^U the restriction vector $(\phi|_{S_{R_1}}, \dots, \phi|_{S_{R_\nu}})$ ranges over the *entire* product $\prod_i D^{S_{R_i}}$, of size $\prod_i N_i$. Deterministic detection supplies for each ϕ a cell $c(\phi)$ with $\phi|_{S_{R_i}} \in T_i(c(\phi))$ for every i , and a fixed cell witnesses at most $\prod_i |T_i(c)|$ distinct restriction vectors. Counting both ways,

$$\begin{aligned} \prod_i N_i &\leq \sum_c \prod_i |T_i(c)| \\ &\leq K \prod_i \beta \rho N_i / K \\ &= K (\beta \rho / K)^\nu \prod_i N_i, \end{aligned}$$

and cancelling $\prod_i N_i$ leaves $\rho \geq K^{1-1/\nu} / \beta$. The bound is proved on one skew-free instance and so bounds the worst case regardless of what other instances look like. Achievability is unconditional: by Proposition F.1, ρ is a function of the share vector and no instance can raise it. **(iii)** is immediate from $\tau_A^* < d$. $\square$

Appendix E. The Certificate

Theorem E.1 (Structural amplification). *Fix a partition Π and let $C_\Pi(e)$ be the cells whose evidence a forged relational item e can alter. Then $|C_\Pi(e)| \leq \rho_\Pi(e) \leq \rho_{\max}$, and for a poison set P ,*

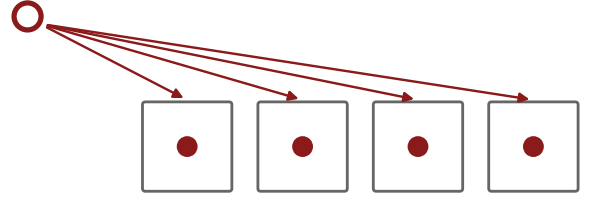
$$\left| \bigcup_{e \in P} C_\Pi(e) \right| \leq \rho_{\max} |P|.$$

Theorem E.1 is where replication stops being an analogy and becomes the attacker's budget conversion rate (Figure 3). Everything after it prices $\rho_{\max}$.

Theorem E.2 (Certificate). *Under A1–A2, for any (m_d, m_s) -Stackelberg adversary,*

$$\mu > m_d + \rho m_s \implies F^{\text{adv}} = F.$$

one relation



every cell it reaches falls together: ρ of K

Figure 3. Why isolation and exposure are the same lever. A cell fires only on a complete witness, so a relation shared by many witnesses must be copied into many cells; one forged copy then lands in every one of them. Detection forces the replication that the adversary converts into reach.

Conversely, under A3 and A5, for every clean instance with $\mu \leq m_d + \rho m_s$ there exists an (m_d, m_s) -adversary that flips the verdict.

Writing $t = \mu$ for the corrupted cells the aggregator tolerates, a deployment quotes

$$b_{\text{cert}} = \left\lfloor \frac{t-1}{\rho_{\max}} \right\rfloor = \Theta(t \cdot K^{-\alpha^*(Q)}). \quad (1)$$

The certificate does not depend on the poison being distinguishable: both Theorems E.1 and E.2 count cells and neither reads content, so a poison that is perfectly plausible, anchored to authentic facts and temporally coherent is bounded by exactly the same $\rho_{\max} m_s$ as a clumsy one. This is what a semantic defense cannot offer, and it is why KEPO's anchoring does not enter the analysis: anchoring changes how likely a poison is to be *retrieved*, which moves m_s , and cannot change $\rho_{\max}$, which is fixed by the partition. Sharing relations across m targets is likewise linear rather than compounding, since a forged relation's cell-set depends on its content alone: it buys the adversary breadth, never depth.

Remark E.3 (Reaching a cell is not the same as flipping it). A5 makes the certificate tight, and it is worth stating what it costs, because the natural reading of the defense contradicts it. A cell fires iff it holds a complete witness, and the adversary may only *add* evidence. Adding tuples cannot destroy a witness, so with an exact subgraph checker and unbounded context a forged relation can never silence a firing cell and no suppression attack exists. Three deployments differ here. With an *LLM reader*, one replicated payload can talk a reader out of a conclusion its context still supports, which is A5 and the setting we assume. With a *capacity-bounded cell* the adversary must evict honest evidence, costing $\Theta(B)$ forged relations per cell, so the certificate is conservative by a factor B . With an *exact checker with room to spare* only the false-positive direction of Section 5 remains. The upper bound is sound in all three; only the matching lower bound picks the first.

Appendix F. The Packing Law: Constants and Skew

The constant, and a warning about fits. A retrieved tuple may fill several atoms at once. Write $\sigma(e)$ for the number of atoms tuple e can fill and $\sigma = \max_e \sigma(e) \leq d$ for the resulting slot constant, used unqualified below. Its cell-set is then a union of slices and $\rho(e) \leq \sigma(e)K^{1-1/\tau_A^*}$, with inclusion–exclusion exact for the anchored C_4 at $\rho_{\max} = d(K^{1-1/\tau_A^*} - 1)$. A naive log–log fit of $\rho_{\max}$ against K reads the -1 as part of the exponent, and on the C_4 that fit returns 0.580 against the true 0.500; the finite- K gains an oblivious design realises carry this $\sigma \leq d$ factor, which the exponent ratios quoted in Section 8 do not.

Proposition F.1 (Skew separation, and why the textbook remedy is unsound). *Fix shares x with $p_u = K^{x_u}$. For every atom R and every instance,*

$$\rho(R) = \prod_{u \notin \text{vars}(R)} p_u,$$

$$\rho(R) = K^{1 - \sum_{u \in \text{vars}(R)} x_u},$$

a function of the share vector alone, so amplification is invariant to the data distribution. Load is not. Moreover $\tau_{A \cup \{u\}}^(Q) \geq \tau_A^*(Q)$: anchoring deletes a packing constraint, so pinning a heavy hitter, the standard skew remedy in one-round parallel joins [43], raises $\rho_{\max}$.*

Measured (SKEW-AND-PINNING): ρ is identical across five skew levels while the max/mean load ratio moves $1.24 \rightarrow 9.49$, and the anchoring inequality holds in 400/400 random queries, strictly in 103. On the triangle at $K = 1024$, pinning one variable moves τ_A^* from $3/2$ to 2 and $\rho_{\max}$ from 10 to 32 : a $3.2\times$ loss of certified budget bought to fix a load problem the certificate never saw. The practical rule inverts: fix skew by *rebalancing buckets*, which cuts the load ratio $2.59 \rightarrow 1.72$ and leaves $\rho_{\max}$ untouched, never by pinning.

Appendix G. What the Adversary Can Reach

Theorem E.2 charges the adversary for every cell a forged relation *occupies*. Its actual power is smaller, for two reasons the construction itself supplies: it cannot choose an arbitrary cell-set, and most of the cells it reaches were not firing.

Remark G.1 (A3 is not reachable under Definition A.2). A content’s cell-set is fixed by the hash vectors of its endpoints: a relation (a, b) fills an atom $R(x, y)$ with $x \leftarrow a$ and $y \leftarrow b$, so its cell-set is the union, over the atoms it can fill in either position, of the slices pinning $h_x(a)$ and $h_y(b)$, and either endpoint may be the anchor. The number of distinct reachable cell-sets is therefore at most $(K+1)^2$, one hash vector per endpoint, against the $\binom{K}{\rho}$ that A3 posits; for the anchored C_4 at $K = 256$, where $\rho_{\max} = 60$, that is at most 66,049 sets against roughly 10^{68} . A3 is sound as a hypothesis for the *upper* bound, which needs only $|C_{\Pi}(e)| \leq \rho_{\max}$, but the

converse half of Theorem E.2 establishes tightness against an adversary this design cannot host.

Definition G.2 (Reachable damage). With F the clean firing set and $C(t)$ the cell-set of content t ,

$$\rho_F := \max_t |C(t) \cap F|,$$

the largest number of *firing* cells one forged relation can enter. Write ρ_R for the size of one slice of atom R .

Proposition G.3 (Sandwich, and computability).

$$\max\left(1, \frac{c_1 \rho_R}{K}\right) \leq \rho_F \leq \min(\sigma \rho_R, c_1),$$

and ρ_F is computed exactly in $O(K^2 d c_1)$ time by enumerating the two endpoint hash vectors and unioning the slice of every atom position the content can fill.

Theorem G.4 (Instance certificate). *Under A1–A2, for any (m_d, m_s) -adversary,*

$$\mu > m_d + \rho_F m_s \implies F^{\text{adv}} = F,$$

never weaker than Theorem E.2 and strictly stronger whenever $\rho_F < \rho_{\max}$.

Measured (REACHABLE-DAMAGE), evaluating Definition G.2 with the adversary free to choose both endpoint vectors and the union over every slot and orientation its content can fill: $\rho_F = \rho_{\max}$ at 18 of 20 anchors on the 2-path at every K , and at 9 of 12 anchors on the anchored C_4 at $K = 16$ and $K = 64$, with the median exactly $\rho_{\max}$ on both motifs,

$$\rho_F = \rho_{\max} = 2 \quad (2\text{-path}),$$

$$\rho_F = \rho_{\max} = 28 \quad (C_4, K = 64).$$

Enumerating a single hash vector, the natural shortcut, reports 7 against 12 and 15 against 28: one row or one column of the cube, where a content reaches a row *and* a column. The firing set is dense enough that some slice-union lies entirely inside it, so the instance certificate is sound but, on this instance, no tighter than Theorem E.2. The security experiments of Section 8 record $|\bigcup_e C(e)|$ against $\rho_{\max} b$ and never intersect the footprint with a firing set, so they confirm Theorem E.1, a statement about occupancy.

Appendix H. The Redundancy Ceiling

The certified budget of (1) is a ratio and Section 9 priced its denominator exactly. Its numerator has been carried as a property of the clean instance, which is how partition-aggregation certificates have always treated it [9], [21]. For relational evidence that is wrong, and in a way that matters.

Proposition H.1 (Margin–share coupling). *Under Definition A.2 every witness ϕ lies in the single cell $c^\phi = (h_u(\phi(u)))_{u \in U}$, so with n_u the number of distinct values u takes across witnesses and W their number,*

$$c_1 = |\{(h_u(\phi(u)))_{u \in U} : \phi\}|,$$

$$c_1 \leq \min \left\{ K, W, \prod_{u \in U} \min(p_u, n_u) \right\}.$$

Coordinates with $a_u = 0$ have $p_u = 1$ and contribute no factor, so the clean margin $\mu = c_1 - \lfloor K/2 \rfloor$ is a function of the share vector and the instance, not of the instance alone.

The bound is not slack where it matters most. The share LP is frequently degenerate, putting *all* mass on the variables shared by the most atoms and none on the rest. For the two-hop entity query

$$Q(ans) := R(v, p), S(ans, p),$$

which has $\tau_A^* = 1$, the optimum is $a_p = 1$, $a_{ans} = 0$, so $c^\phi = h_p(\phi(p))$ and $c_1 \leq n_p$, the number of distinct intermediate entities; for the anchored 2-path the same shares give $c_1 \leq \deg(v)$. **The queries whose replication the packing law drives to 1 are exactly those whose firing set it drives to a handful of cells.**

Corollary H.2 (When majority is vacuous). *The certificate is non-trivial only if $K < 2c_1 \leq 2W$. For $K \geq 2c_1$ the clean verdict is $F = 0$: the defense reports the pattern absent on an instance where it is present, and no budget is certified at any ρ .*

Majority is inherited from defenses in which every group is a meaningful voter because every group holds a self-contained passage [21], [9]. Here a cell holding no witness is not a dissenting voter but a vacuous one, and there can be many more of them than of the informative kind. Thresholding on the firing count costs nothing and removes the requirement.

Theorem H.3 (Recentred certificate). *Let each cell report $\hat{h}_i = h_i \oplus \xi_i$ with per-cell error rate $\varepsilon' < 1/2$ and correlation budget κ , fix $\theta \in (0, K)$, and report $\hat{F} = 1$ iff $\hat{X} \geq \varepsilon' K + (1 - 2\varepsilon')\theta$. With*

$$\begin{aligned} \Delta &= m_d + \rho m_s, \\ \Psi &= \frac{1}{1 - 2\varepsilon'} \sqrt{\frac{\kappa K \ln(1/\delta)}{2}}, \end{aligned}$$

each direction holds with probability at least $1 - \delta$: suppression is safe if $c_1 - \theta > \Delta + \Psi$, and forgery is safe if $\theta > \Delta + \Psi$. Both are satisfiable iff $c_1 > 2\Delta + 2\Psi$, and $\theta = c_1/2$ balances them, giving

$$b_{\text{cert}} = \left\lfloor \frac{c_1/2 - \Psi - m_d}{\rho_{\max}} \right\rfloor.$$

Setting $\theta = K/2$ recovers the majority certificate exactly.

Two things are gained: the guarantee becomes *two-sided*, where Theorem E.2 bounds only suppression, and the requirement on the instance weakens from $c_1 = \Omega(K)$ to

$$K < \frac{c_1^2 (1 - 2\varepsilon')^2}{2\kappa \ln(1/\delta)},$$

a quadratic improvement in how much redundancy a given K demands.

Corollary H.4 (Redundancy ceiling and the optimal cell count). *For every K , every share vector and every $\rho_{\max} \geq 1$,*

$$b_{\text{cert}} \leq \frac{c_1}{2\rho_{\max}} \leq \frac{c_1}{2}.$$

No majority or recentred certificate certifies more forged relations than half the number of cells its clean evidence can light up, however many cells it buys.

Corollary H.5 (The optimal number of cells). *Since Proposition G.3 lower-bounds one poison's damage by $\max(1, c_1 \rho_R / K)$, applied to the atom with the largest slice, for which $\rho_R = K^{1-1/\tau_A^*}$,*

$$b_{\text{cert}} \leq \frac{1}{2} \min(c_1, K/\rho_R),$$

$$\text{that is } b_{\text{cert}} \leq \frac{1}{2} \min(c_1, K^{1/\tau_A^*}),$$

whose two terms meet at

$$K^* = c_1^{\tau_A^*},$$

$$\text{at which } b_{\text{cert}} = \Theta(c_1).$$

Buying cells beyond K^ buys nothing while the noise fee $\Psi = \Theta(\sqrt{K})$ keeps growing. So cells help, but only until the evidence runs out; and τ_A^* changes role, becoming the exponent of the compute an oblivious partition needs to reach the redundancy-limited optimum rather than a tax on the robustness it can certify.*

For passages $\rho = 1$ and every group votes, so $K^* = c_1$ [21]; for relational evidence under a content-oblivious partition the same budget costs $K^* = c_1^{\tau_A^*}$ cells. Three smaller consequences of the same LP are stated and proved in Appendix O.

Appendix I. Escapes, in Brief

Three idealisations a deployment cannot honour are priced in Appendix Q. Private randomness blinds an attacker who must *invent* relations, by $\Theta(K^{1/\tau_A^*})$ at the detection threshold and by at most $K/c_1 \leq 2$ with a margin; it does not blind one who *copies* an honest relation with a poisoned provenance document, since A1 routes the copy exactly where the original is. Adaptivity does not erode the first, which is why the interface returns a verdict and never per-cell scores; an imperfect per-cell verdict costs a second-order $\sqrt{K}$ fee that survives correlated reader errors at a $\sqrt{\kappa}$ premium. Under the witness partition both questions are moot: a copy lands in at most one selected witness, and $\kappa = 1$.

Appendix J. Measurements of the Packing Law and the Attack Calculus

Everything is CPU-only: LP, combinatorics and Monte Carlo over integer sets. A reproduction gate asserts every load-bearing number against committed results and exits non-zero on drift. Each experiment is named for the quantity

it measures; `experiments/EXPERIMENTS.md` maps every name to the script and results file that reproduce it. Five adversaries are code, not assumptions: ADV-WB (white-box greedy), ADV-BLIND (private partition), ADV-ADAPT (reads the 1-bit verdict and retracts misses), ADV-SLICE (reachable cell-sets are coordinate slices) and ADV-SKEW (targets the highest-replication positions).

The law itself. The share LP value equals $1/\tau_A^*$ with duality gap below 10^{-15} on 15 motif families, the anchored HyperCube attains $P_{\text{det}} = 1$ at every K , motif and anchor, and its measured amplification exponent matches $1 - 1/\tau_A^*$ to within 0.001.

ALL-PATTERNS: the law is not about cycles. Because the running examples are cycles, it is fair to ask whether the gain is an artifact of the shapes we chose. ALL-PATTERNS removes the choice: enumerate every connected pattern with at most 5 edges up to isomorphism (22 shapes), anchor at every vertex orbit (58 shape–anchor pairs), and compute τ_A^* . It is strictly below d in 53/58 cases, with a median certified-budget gain of $3.17\times$ and up to $256\times$ at $K = 1024$. Cycles are mid-table: broken out by family the median gains are trees $5.66\times$, non-simple cyclic patterns $4.00\times$, cycles and paths $3.17\times$, stars $1.00\times$. The five exceptions are exactly the stars anchored at their centre, which are matchings on their free variables, so Theorem A.3(iii) predicts equality; anchoring the same star at a leaf instead sends τ_A^* from 4 to 1. Monotonicity holds in 600/600 random queries, strictly in 110.

PEGASE: a real network, and the margin it hides.

We run the design on the PEGASE pan-European transmission topology [7] shipped with `pandapower` ($n = 2869$, $m = 3968$, mean degree 2.77, max 15), anchoring at 38 buses (Table 6 in the appendix). Detection is deterministic at every anchor and every K ; the 2-path needs no replication while the arity floor climbs $4 \rightarrow 8 \rightarrow 16$; and for C_4 the amplification matches the exact slot formula $d(\sqrt{K} - 1)$ to the integer. Replacing uniform hashing with frequency-balanced bucketing cuts the mean max/mean cell-load ratio from 2.59 to **1.72** at $K = 256$ while leaving ρ_{max} identical, so load skew and poison amplification are different quantities with different fixes.

What PEGASE does not report is c_1 , and by Proposition H.1 that omission is load-bearing. For the 2-path the share LP returns $(a_x, a_y) = (1, 0)$, so a witness’s cell is $h_x(x)$ with x a neighbour of the anchor, giving $c_1 \leq \deg(v) \leq 15$ at every K . Majority aggregation therefore gives

$$\mu \leq 15 - 32 = -17 \quad (K = 64),$$

$$\mu \leq 15 - 128 = -113 \quad (K = 256),$$

at every anchor: the rows of Table 6 are collected in the regime Corollary H.2 rules out. P_{det} and ρ_{max} are exactly as predicted, and the aggregator they feed certifies nothing.

The attack calculus against implemented adversaries. Five adversaries are code, not assumptions, and every prediction of Section E holds against them. Appendix S gives the measurements: the certified success bound is never violated, blinding buys at most the theorem’s ceiling with a margin

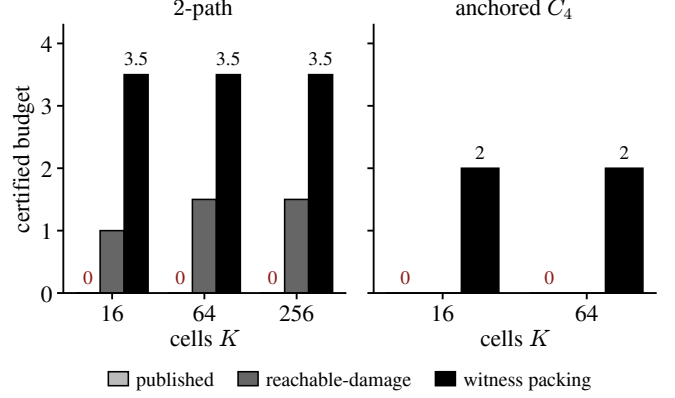


Figure 4. THREE-DESIGNS on PEGASE. The majority certificate is zero at every K and both motifs; charging only reachable cells recovers a little on the 2-path and nothing on the C_4 ; isolating witnesses recovers 3.5 and 2 from 8 and 6 cells respectively, against the hundreds the oblivious design buys.

and a polynomial factor at the detection threshold, adaptivity buys one percent, and certifying with the minimum of the two tails beats either alone in every configuration. The same appendix reports correlated readers end to end (89/89 settings certified; a control at a quarter margin breaks 154/200) and that amplification is linear in the number of targets.

REACHABLE-DAMAGE, THREE-DESIGNS and DUPLICATE-ATTACK. `witness_packing.py` runs on the same PEGASE topology, recording the host in its results file, and selects the witness family by an exact set-packing program (Remark 6.2); greedy attains the maximum at 20 of 20 2-path anchors and 10 of 12 C_4 anchors. The majority certificate of Theorem E.2 is vacuous at every K and at 19 of 20 anchors already at $K = 16$; charging only reachable cells recovers a median of 1.5 on the 2-path and nothing on the C_4 , where $\rho_F = \rho_{\text{max}}$; while the witness partition certifies a median of 3.5 forged relations using 8 cells on the 2-path, and on the anchored C_4 it certifies 2 using 6 cells (Figure 4) — more budget from an order of magnitude fewer cells. The same run prices the duplicate attack of Appendix Q: a copy of the honest relation lying in the most witnesses, chosen from the graph alone, reaches a median $\rho_F/2$ of the firing cells on both motifs (0.50 on the 2-path, 0.54–0.58 on the C_4), never fewer than one at any anchor, and on the 2-path at $K = 256$ that is $16\times$ what a fresh content reaches in expectation under a private hash.

Appendix K. Proof of Theorem E.2

Each forged document enters one cell (A1); each forged relation enters at most ρ_{max} (Theorem E.1), so $|X^{\text{adv}} - X| \leq m_d + \rho m_s$ and $\mu > m_d + \rho m_s$ forces $X^{\text{adv}} > \lfloor K/2 \rfloor$. For the converse, WLOG $F = 1$; the adversary routes m_d documents via A3 to distinct firing cells and, for each of m_s forged relations, targets a ρ -subset of still-firing cells, flipping each by A5. Total silenced $\geq m_d + \rho m_s \geq \mu$. $\square$

TABLE 5. EVERY LOAD-BEARING HYPOTHESIS. TWELVE FAIL, AND ALL TWELVE ARE PREDICTIONS THIS FRAMEWORK MADE. A FRAMEWORK THAT CANNOT FALSIFY ITS OWN PREDICTIONS IS NOT BEING TESTED.

Hypothesis	Verdict	Evidence
share LP value = $1/\tau_A^*$	holds	duality gap $< 10^{-15}$, 15 families
anchored HyperCube meets K^{1-1/τ_A^*}	holds	exponent to 4 d.p., $P_{\text{det}}=1$
$\nu = \tau_A^*$, so the native bound is tight	holds	35/35 families and TPC-H queries
$\tau^* < d$ on a real workload	holds	TPC-H: 18/20, gap $\leq 5.6 \times 10^{-17}$
two-table FK joins need $\rho = 1$	holds	TPC-H: $\tau^*=1$ on all 8
$P_{\text{det}}=1$ on a real network	holds	PEGASE: PEGASE, all anchors and K
ρ invariant to the data distribution	holds	SKEW-AND-PINNING: identical at 5 skew levels
anchoring never lowers τ_A^*	holds	SKEW-AND-PINNING: 400/400, strict in 103
certified success bound never violated	holds	TAIL-BOUND: 0/50
min(Bernstein, Hoeffding) beats either	holds	BERNSTEIN-GAIN: 18/18, mean +4.6%
blinding buys $\Theta(K^{1/\tau_A^*})$ at onset, fresh contents	holds	BLINDING-GAIN: fits 0.500, 0.343
adaptivity buys $\leq$ a constant	holds	ADAPTIVITY: 1.011 $\times$
$\kappa \leq \rho m_s$ needs no measurement	holds	CORRELATION-CHAINING: 36/36
end-to-end under correlated failure	holds	END-TO-END: 89/89; control breaks 154/200
witness packing gives $\rho = 1$	holds	Thm 5.2; THREE-DESIGNS
$\omega \leq \nu^* \leq \tau$ on real evidence	holds	FRACTIONAL-GAIN: $\nu^*/\omega \leq 1.50$, $\tau/\nu^* \leq 1.85$
transversal rule beats witness packing	holds	FRACTIONAL-GAIN: C_4 median 5 against 2
a declared class makes claims certifiable	holds	CLAIM-CHECK: 72.3% of claims, 11.7% of non-claims
budget of a two-hop answer is an edge cut	holds	FLOW-CHECK: 464/464 answers
the sparse regime of FLOW-CHECK is the general case	holds	METAQA: 96.4%/92.5% single-chain; HETIONET: 63.5–93.7%
$K^{1-1/d}$ is a floor on <i>any</i> design	fails	HyperCube hits K^{1-1/τ_v^*} , $P_{\text{det}}=1$
co-location is the one-round load problem	fails	Rem. T.1: $\rho_A^* < \tau_A^*$ on C_3, C_5, Θ
A3 is realisable under the deployed design	fails	Rem. G.1: $\leq (K+1)^2$ reachable cell-sets
ρ_F is well below ρ_{max} on real instances	fails	REACHABLE-DAMAGE: exact $\rho_F = \rho_{\text{max}}$ at 18/20 and 9/12 anchors
a private hash blinds every structural attack	fails	DUPLICATE-ATTACK: a copy reaches $\rho_F/2$, ≥ 1 at every anchor
the margin is a property of the instance	fails	Prop. H.1: it follows the shares
majority is the right aggregator	fails	Cor. H.2: vacuous on PEGASE
more cells always help	fails	Cor. H.5: optimum at K^*
$\rho = 1$ needs a matching query	fails	Thm 5.2: it needs witness cells
$\kappa \leq \rho$ (one payload's multiplicity)	fails	CORRELATION-CHAINING: chains to 7.6ρ at $m_s=8$
the limit is half the fractional packing	fails	Thm 5.4: it is the transversal τ
witness cells defend the full answer margin	fails	Cor. 6.3: halved at ≥ 3 hops

Appendix L.

Proof of Proposition G.3 and Theorem G.4

The slices of one atom partition $[K]$ into K/ρ_R parts whose intersections with F sum to c_1 , so some slice meets F in at least the mean $c_1\rho_R/K$; a content reaching no firing cell does no damage, giving the 1. The upper bound is $|C(t) \cap F| \leq \min(|C(t)|, |F|)$. Enumerating the endpoint pair (g_a, g_b) over $(\prod_u [p_u] \cup \{v\})^2$ and unioning the slice of every atom position the content fills evaluates the maximum directly, in $O(K^2 dc_1)$; a single vector under-counts whenever a content fills two positions, which on an undirected host every content does. For the certificate, only a firing cell can be silenced and each forged relation enters at most ρ_F of them, so the counting of Theorem E.2 applies verbatim with ρ_F in place of ρ . $\square$

Appendix M.

Proof of Proposition H.1 and Theorem H.3

The cell of a witness is computed in the proof of Theorem A.3(i): every atom's tuple agrees with ϕ on its free variables, hence lies in c^ϕ . The map $\phi \mapsto c^\phi$ factors through $(h_u(\phi(u)))_u$, whose u -th coordinate takes at most $\min(p_u, n_u)$ values, giving the product bound.

For the recentred certificate, the reported count has mean $\mathbb{E}[\hat{X}] = \varepsilon'K + (1 - 2\varepsilon')c'_1$ with c'_1 the post-attack firing count. In the suppression direction $c'_1 \geq c_1 - \Delta$, so

$$\mathbb{E}[\hat{X}] - \varepsilon'K - (1 - 2\varepsilon')\theta \geq (1 - 2\varepsilon')(c_1 - \Delta - \theta),$$

and Hoeffding over the correlation groups, whose sizes satisfy $\sum_g |g|^2 \leq \kappa K$, bounds the lower deviation by

$$\exp(-2(1 - 2\varepsilon')^2(c_1 - \Delta - \theta)^2/(\kappa K)) \leq \delta$$

under the stated inequality. The forgery direction is the same computation with $c'_1 \leq \Delta$ and an upward deviation, since a forged item lights a cell only if it reaches it. Adding the two conditions gives $c_1 > 2\Delta + 2\Psi$, with the two slacks equal at $\theta = c_1/2$. At $\theta = K/2$ the threshold is $\varepsilon'K + (1 - 2\varepsilon')K/2 = K/2$, the majority rule. $\square$

Appendix N.

Proof of Proposition F.1

Hashing sends a tuple of R to one bucket in each coordinate of $\text{vars}(R)$ and to all p_u buckets in each coordinate it does not constrain, so it reaches $\prod_{u \notin \text{vars}(R)} p_u$ cells, independent of the values it carries and hence of the instance. Moving u from free to anchored deletes the row $\sum_{R \ni u} x_R \leq 1$ from

the packing LP; deleting a constraint relaxes a maximisation, so τ_A^* cannot decrease and K^{1-1/τ_A^*} cannot fall. On the triangle, free, the optimum is $x_R \equiv 1/2$ with value $3/2$; with one variable anchored the two atoms containing it are unconstrained and the optimum is 2. $\square$

Appendix O. Three Consequences of the Share LP

Shares beyond a variable’s witness diversity buy no cells, so the design problem is the share LP under capacity constraints $a_u \leq \log_K n_u$ with objective $c_1(a)/\rho_{\max}(a)$, whose optimum is not the replication-optimal one: on the anchored 2-path the LP optimum certifies nothing where an even split certifies a positive budget. The correlation budget is geometry rather than a free parameter, since a tuple’s cell-set is a coordinate sub-cube and sub-cubes fixing different coordinates always intersect, so $\kappa \leq \rho$ when every forged tuple fills atoms of one support and $\kappa \leq \min(K, \rho m_s)$ otherwise. And giving the *anchor* a share never helps: it restricts firing to a slab of K^{1-a_v} cells while multiplying the replication of every atom that does not contain the anchor by K^{a_v} , which closes the exit that Proposition F.1 appears to open.

Appendix P. The Blind Attack and the Two Channels

The total corpus surface is $S = \sum_e \rho(e) = K\bar{B}$ by double-counting the incidence set $\{(e, c) : e \in c\}$: the attack surface is *exactly* the context the defender pays the verdict function to read, so buying cells and reducing exposure are one lever. Under A3 the white-box adversary silences $\rho_{\max}$ firing cells per poison and needs exactly $\lceil(\mu - m_d)/\rho_{\max}\rceil$. Under a private partition a fresh content’s cell-set is uniform, a fixed firing cell survives with probability exactly

$$q(m_d, m_s) = \left(1 - \frac{1}{K}\right)^{m_d} \left(1 - \frac{\rho}{K}\right)^{m_s},$$

and the attack becomes likely once $c_1 q \leq \lfloor K/2 \rfloor$, i.e. once $m_s = \Theta(K/\rho) = \Theta(K^{1/\tau_A^*})$. The success probability of a blind attack of given size admits a *certified* bound rather than a heuristic one, because the survival indicators are negatively associated [24], [29], so Hoeffding and Bernstein both apply and a defender certifies with their minimum; the mean-field binomial law is accurate but two-sided and we report it as an estimate. Finally $\rho_{\max}$ is an exchange rate between the two attack channels: an all-structural attack is cheaper than an all-document one iff

$$c_s/c_d < \mu/\lceil\mu/\rho_{\max}\rceil \leq \rho_{\max},$$

so lowering ρ does not merely raise the price of structural poisoning, below the threshold it pushes the adversary onto the document channel where the certificate is ρ -free.

Appendix Q. Escapes: Randomness, Adaptivity, Noise

Private randomness (A4), re-drawn per query, multiplies the budget of a blind attacker who must *invent* relations by $\Theta(K^{1/\tau_A^*})$ at the detection threshold, where a white-box adversary needs $O(1)$ poisons, and by at most $K/c_1 \leq 2$ in the majority regime. It is no exit against the *duplicate attack*: a forged copy of an honest relation e with a poisoned provenance document is routed by A1 exactly as e , so its cell-set is $C(e)$ whatever the hash, and $C(e)$ contains the cell of every witness through e , all of which fire; DUPLICATE-ATTACK measures it in Section 8. *Adaptivity* does not erode the first gain: an adversary that reads the one-bit verdict and retracts misses still pays $\Omega(K/\rho)$ injections, because a fresh content’s cell-set is uniform. If the interface leaks b bits per query rather than one, the decision-tree bound gives

$$\Pr[\text{success}] \leq 2^{Tb} \Pr[\text{blind success}],$$

so the defense tolerates only $T = O(\log_2 K/db)$ rich queries: **return the verdict, not the scores**. *Noisy verdicts* cost $\sqrt{K \ln(1/\delta)/2}/(1 - 2\epsilon')$ with independent errors and degrade as $\sqrt{\kappa}$ under a correlation budget (Theorem H.3); the threat model bounds $\kappa \leq \min(K, \rho m_s)$, so soundness needs no measurement, and under Theorem 5.2(v) the question disappears.

Appendix R. Certifying in the Attacker’s Units

Published structural attacks budget in injected *documents*, and the map to relations is the extractor’s yield c_{edge} , which Section 3 declines to bound. Building H over provenance documents instead, a witness is the set of documents its relations came from, and Theorem 5.4 applies verbatim with $\tau_{\text{doc}} \leq \tau$: the certificate is then stated in forged documents, the units the attacks are measured in.

Appendix S. Attack-Calculus Measurements

Over 50 configurations spanning $K \in \{64, 256\}$, $\tau_A^* \in \{2, 3\}$ and budgets to $4\times$ the blind threshold, the certified success-probability bound is **never violated** (0/50) while the mean-field estimate tracks measurement to a mean absolute error of 0.012 (max 0.119, signed range $[-0.079, +0.067]$), which is why we certify with the first and forecast with the second. The surface identity $S = K\bar{B}$ holds with gap exactly 0 at every anchor and every K on PEGASE ($n = 2869$), and the surface is not concentrated: the single highest-replication edge carries a mean 3.3% of it at $K = 16$ and 3.0% at $K = 64$, so no single edge’s removal collapses the adversary’s options. In the margin-rich regime blinding buys between $1.2\times$ and $2.0\times$, always under the theorem’s ceiling $K/c_1 \leq 2$. At the detection threshold it buys a polynomial factor: the blind budget is 8.01 ± 0.12 at $K =$

64, 15.77 ± 0.24 at $K = 256$, 32.08 ± 0.49 at $K = 1024$ and 63.59 ± 0.96 at $K = 4096$, a separation of **63.6** $\times$, with fitted exponents **0.500** against the predicted $1/\tau_A^* = 0.500$ and 0.343 against 0.333; the blind budget is exactly geometric with success probability ρ/K , and measurement agrees with its mean K/ρ at $|z| < 3$ in all eight settings (max $|z| = 2.92$). Adaptivity buys 1%: ADV-ADAPT’s budget is **1.011** $\times$ that of the non-adaptive blind attacker, exactly as the supermartingale argument predicts, because a 1-bit verdict against a private partition reveals only whether *this* content hit. Certifying with min(Bernstein, Hoeffding) beats Hoeffding alone in 18/18 configurations, buying a mean 4.6% and up to **50%** more certified budget at the same confidence. Finally the slice control: for a generic firing set the deployed partition matches the idealised uniform- ρ -subset model to within $[0.983, 1.042]$ of required budget with variance inflation **1.62** $\times$, while the 31.7 $\times$ inflation a naive measurement reports needs F aligned with a hash coordinate, which an adversary can arrange only by seeing the hash.

Appendix T. Co-location Is Not the One-Round Load Problem

Remark T.1 (Co-location is not the one-round load problem). Importing the load bound of [3] needs two steps: a cell holding at most B tuples per atom contains at most $B\tau_A^*$ witnesses, and a hard instance realises $N\tau_A^*$ witnesses. Neither survives anchoring. The witnesses a cell can hold are governed by the fractional edge cover ρ_A^* of the free variables, by the AGM bound [2]: anchored scatter-gather with $b = 3$ has $\tau_A^* = 2$ against $\rho_A^* = 3$, so one cell can hold B^3 witnesses. Conversely AGM caps the witness count at $N\rho_A^*$, and $\rho_A^* < \tau_A^*$ for the anchored triangle (1 against 2), C_5 and the theta graph (2 against 3); anchoring also collapses the matching database of [3, Lemma 4.5]. The cause is single: a cell must hold a *whole* witness, not enough of one to contribute to an aggregate, so co-location is strictly stronger than one-round evaluation. Theorem A.3(ii) is therefore proved directly, and a packing is exactly a set of atoms whose evidence varies independently.

Appendix U. Correlated Readers, Amplification and Tightness

Correlated readers, end to end. The component bound $\kappa \leq \rho m_s$ holds in 36/36 configurations of (K, ρ, m_s) ; chaining does push κ well above ρ , to 7.6ρ at $m_s = 8$, which is why the single-payload reading is unsafe; the certified-regime inequality $\kappa \leq \rho m_s < \mu \leq K$ holds in all 1548 random draws; and block-Hoeffding is never violated across 20 tail points. Tested end to end rather than tail by tail, running the worst-case adversary and then flipping verdicts with whole overlap-components failing together, the certificate holds in

TABLE 6. PEGASE. ANCHORED HYPERCUBE ON PEGASE. DETECTION IS DETERMINISTIC EVERYWHERE; FOR THE 2-PATH REPLICATION IS FLAT WHILE THE ARITY FLOOR GROWS, AND FOR C_4 IT MATCHES $d(\sqrt{K} - 1)$ TO THE INTEGER.

motif	K	P_{det}	ρ_{max}	arity	skew	bal.
2-path (2, 1)	16	1.000	2	4.00	2.57	—
	64	1.000	2	8.00	3.09	—
	256	1.000	2	16.00	3.15	—
C_4 (4, 2)	16	1.000	12	8.00	1.39	1.34
	64	1.000	28	22.63	1.85	1.53
	256	1.000	60	64.00	2.59	1.72

89/89 settings with empirical failure at least 0.01 below δ everywhere. The test has power: at a quarter of the certified margin the same pipeline breaks the verdict in 154 of 200 settings.

Amplification, multi-target and tightness. For every forged tuple the observed $\rho_{\text{obs}}(e) = |C(e)|$ meets the bound: over 12 configurations the ratio is **1.000** everywhere, never exceeded and always attained. Connecting targets does not compound: per-target amplification stays at **16** for every $m \in \{1, 2, 4, 8, 16\}$ while Γ grows with slope **1.00**. Across 20 configurations it reaches $b\rho_{\text{max}}$ and never exceeds it, so knowing the defense completely does not help. Comparing the certificate against the smallest budget at which that attacker actually silences the margin gives a tightness ratio in **[1.02, 1.33]**. As Remark G.1 explains, all four measure occupancy rather than damage.

Appendix V. The PEGASE Measurements

Appendix W. Two Remarks on the Witness Partition

Remark W.1 (Randomising the selection buys nothing). A forged witness edge-disjoint from every other lies in *every* maximum and every maximal packing, so its damage is one cell whatever distribution the selection is drawn from: Theorem 5.2 is deterministic and hiding the partition cannot improve it.

Remark W.2 (Small ω needs exact tails, not Hoeffding). With ω cells and per-cell error ε' , the Hoeffding fee of Theorem H.3 is non-vacuous only when $\omega > 2 \ln(1/\delta)/(1 - 2\varepsilon')^2$, about 10 at $\varepsilon' = 0.1$ and $\delta = 0.05$, and measured ω is often smaller. The cells are independent by Theorem 5.2(v), so the exact binomial tail applies, costs $O(\omega)$, and certifies where the Chernoff form is empty.

Disclosure of LLM Use

In accordance with IEEE policy: a large language model was used for drafting assistance, prose editing, and to help implement and debug the experiment scripts. All theorem statements and proofs were derived and checked by the

authors; every quantitative claim is asserted by the artifact’s reproduction gate against committed data, and the gate fails on drift. The authors take full responsibility for the content.

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